MACROECONOMICS

MACRO-20

Time: 5 minutes

Outcome: Analyze reserves, fractional-reserve banking, balance sheets, multipliers, and deposit creation.

Difficulty: ●●○ Intermediate

Bank Money Creation

THE CORE IDEA

Fractional-reserve banks hold only part of deposits as reserves and can create deposits when they make loans. A new loan typically becomes a deposit elsewhere in the banking system, allowing the process to continue. The simple deposit multiplier describes a maximum expansion when banks lend all excess reserves and the public redeposits the proceeds; excess reserves and cash holding make actual expansion smaller.

HOW TO RECOGNIZE IT

- A bank receives a new deposit and holds all of it as excess reserves instead of lending.
- Deposits are liabilities: the bank has an obligation to repay depositors.
- Borrowers owe repayment to the bank, so loans are bank assets.
- The simple multiplier is 1 divided by the reserve ratio.
- The reserve ratio is the reciprocal of the multiplier: 1/4 = 25 percent.

WATCH OUT

A common mistake is to reverse the reserve-ratio relationship. The simple multiplier is 1 divided by the required reserve ratio, so a higher ratio produces a smaller potential deposit expansion.

WORKED EXAMPLE: DEPOSIT CREATION IN ACTION

A bank has $900,000 in deposits and faces a 12% reserve requirement. Required reserves equal 0.12 x $900,000 = $108,000. If the bank holds exactly that amount, the remaining $792,000 can support loans. The simple money multiplier is 1/0.12 = 8.33, an upper bound rather than a guarantee.

CHECK YOURSELF

When a bank makes a loan from excess reserves, what can happen next?

READY? **Return to the game and master this concept.**